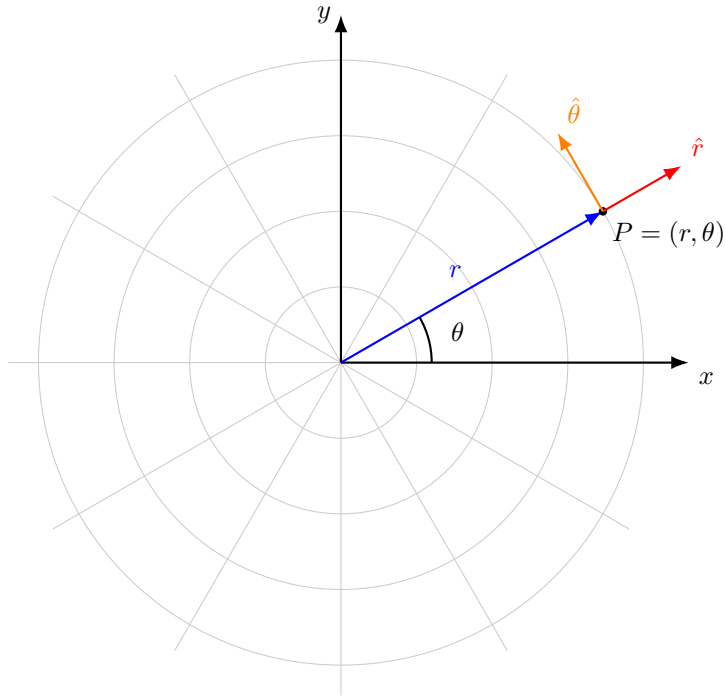


Polar Coordinates

Robert O'Connell

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Example Plot



From the plot we can see that

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Basis Vectors $(\hat{r}, \hat{\theta})$

The basis vectors $\hat{r}$ and $\hat{\theta}$ rotate as θ changes

$\hat{r}$ points radially outward from the origin, while $\hat{\theta}$ is tangent to the circle in the direction of increasing θ

And,

$$\hat{r} \cdot \hat{\theta} = 0$$

$$\Rightarrow \hat{r} \perp \hat{\theta}$$

Defining $(\hat{r}, \hat{\theta})$ from cartesian basis vectors

$$\hat{r} = \cos(\theta)\hat{i} + \sin(\theta)\hat{j}$$

$$\begin{aligned}\hat{\theta} &= -\cos\left(\frac{\pi}{2} - \theta\right)\hat{i} + \sin\left(\frac{\pi}{2} - \theta\right)\hat{j} \\ &= -\sin(\theta)\hat{i} + \cos(\theta)\hat{j}\end{aligned}$$

Position, Velocity and Acceleration Derivatives

$$\begin{aligned}\vec{r}(t) &= r \cos(\theta)\hat{i} + r \sin(\theta)\hat{j} \\ &= r\hat{r}\end{aligned}$$

$$\begin{aligned}\vec{v}(t) &= \frac{d}{dt}(r\hat{r}) \\ &= \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt} \\ \frac{d\hat{r}}{dt} &= -\sin(\theta)\frac{d\theta}{dt}\hat{i} + \cos(\theta)\frac{d\theta}{dt}\hat{j} = \frac{d\theta}{dt}\hat{\theta} \\ \Rightarrow \vec{v}(t) &= \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}\end{aligned}$$

The velocity has two components, one along $\hat{r}$ (changing the distance from the origin) and one along $\hat{\theta}$ (changing the angle)

$$\begin{aligned}\vec{a}(t) &= \frac{d}{dt}(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}) \\ &= \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \dot{r}\dot{\theta}\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt} \\ \frac{d\hat{\theta}}{dt} &= \frac{d}{dt}(-\sin(\theta)\hat{i} + \cos(\theta)\hat{j}) = -\dot{\theta}\hat{r} \\ \Rightarrow \vec{a}(t) &= (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}\end{aligned}$$

The first term, $(\ddot{r} - r\dot{\theta}^2)\hat{r}$, represents the radial acceleration. The negative component $-r\dot{\theta}^2\hat{r}$ corresponds to the centripetal acceleration directed toward the origin.

The second term, $(2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$, represents the tangential acceleration. Here, $2\dot{r}\dot{\theta}$ is the Coriolis component, and $r\ddot{\theta}$ arises from angular acceleration.